

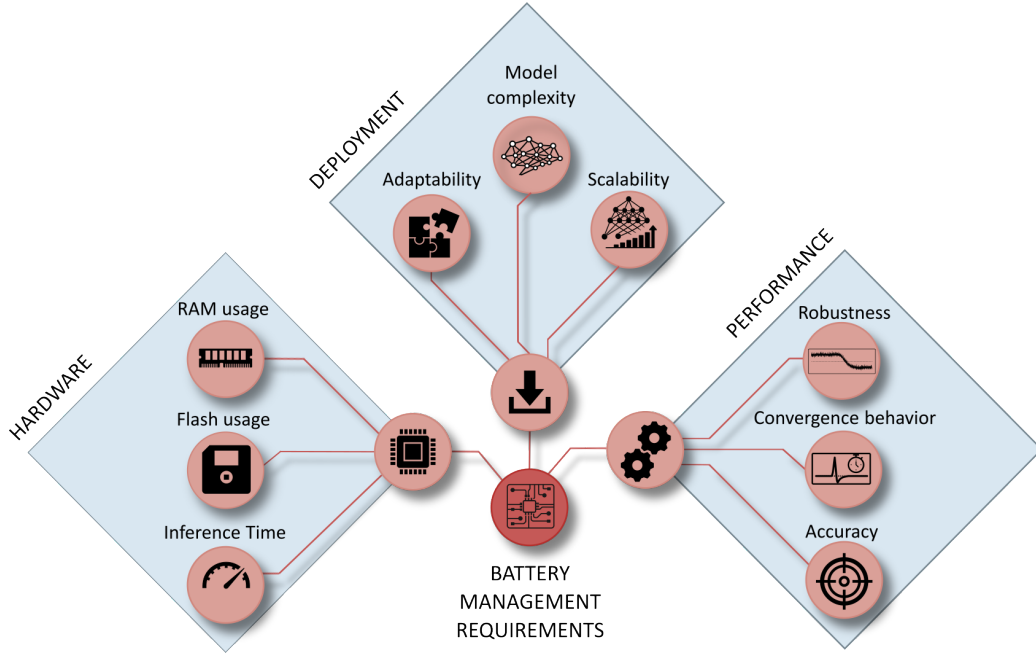


Figure 2.1: Battery-management requirement framework used throughout this work. The three connected dimensions cover estimator performance, the hardware resource envelope, and the engineering path to deployment.

2.1 Performance requirements

Performance requirements describe whether the estimated SOC or SOH signal is suitable for BMS decisions during operation. In this work, performance is not reduced to average error under clean measurements. It also includes robustness against disturbed input signals and convergence or recovery after an inconsistent initial state. This distinction is important because bias, noise, missing samples, timing irregularities, and communication interruptions can change the estimator ranking even when the clean-data error is low.[7, 8, 9]

2.2 Hardware requirements

Hardware requirements define the resource envelope in which state estimation can run on the BMS controller. Flash memory limits the amount of code, parameters, and lookup data that can be stored. RAM usage limits the size of recurrent states, intermediate buffers, stack, and runtime data. Inference time determines whether the estimator can meet the required update rate with sufficient timing margin.